
PH102: Physics-2

Electrodynamics

Jan-May 2024

Lecture # 01
Vector Algebra
04.01.2024

Instruction Details

Instructors

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Professor Ashwini K Sharma



Vector Calculus and
Electrostatics

Professor T. Subhash
Professor Pankaj K Mishra



Electrodynamics,
Magnetostatics and
EM waves

PH : 102 Instructors



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Division 3 (CSE+EPH+MC): Afternoon

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Class Room # (L4)
Division 2 (ECE+EEE+ BT): Morning
Division 4 (CL+CST+DD+DS&AI): Afternoon

PH102

2024

3

Tutorial Team

Group	Faculty	Student TA	Room No.	No. Student
T1	Aksh	Ayan Chakraborty	1103	50
T2	ST	Sreetama Das Choudhary	1201	50
T3	PKM	Swati Shaw	1202	50
T4	Pamu	Nidhi Rawat	1205	50
T5	Arunansu	Vaibhav Pandey	2101	50
T6	Bibhas	Bhoomika Das	2201	50
T7	Bipul	Harpreet Singh	2202	50
T8	Girish	Vishal	2203	50
T9	MKN	Dipankar	2204	50
T10	MCK	Utsab Dey	5001	50
T11	Perumal	Heramb Vivek Bhusane	5002	49
T12	Poulose	Shailesh Pincha	5003	49
T13	Saikat	Nayan Das	5101	49
T14	Santa	Indrajit Saha	5104	49
T15	Sayan	Nihar Ranjan Ghosh	5106	49
T16	Ravi	Gunjan Yadav	5205	49
T17	Soumitra	Hrisikesh Thakur	5206	49
T18	Subhaditya	Camelia Jana	5G2	49
T19	Subhradip	Pooja Kumari Gupta	L2	49
T20	Udit	Jitumani Kalita	L4	16+Backlog student

Assessments

Events	Date	Time	Marks
Quiz-1	06/02/24	45min	15
Mid-Sem			30
Quiz-2	09/04/2024	45min	15
End Sem			40

Lecture Notes and Tutorial Problems can be found at

https://teams.microsoft.com/l/team/19%3av3G1oEUEHnpQk9VLnEo37zBx_BBCFiSxIFi3wplamc1%40thread.tacv2/conversations?groupId=f585b065-4be8-4942-9c42-f872f70fb91d&tenantId=850aa78d-94e1-4bc6-9cf3-8c11b530701c

Syllabus

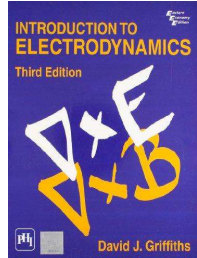
- **Vector analysis:** Gradient, Divergence and Curl; Line, Surface, and Volume integrals; Gauss's divergence theorem and Stokes' theorem in Cartesian; Spherical polar and Cylindrical polar coordinates; Dirac Delta function.
- **Electrostatics:** Gauss's law and its applications, Divergence and Curl of Electrostatic fields, Electrostatic Potential, Boundary conditions, Work and Energy, Conductors, Capacitors, Laplace's equation, Method of images, Boundary value problems in Cartesian Coordinate Systems, Dielectrics, Polarisation, Bound Charges, Electric displacement, Boundary conditions in dielectrics, Energy in dielectrics, Forces on dielectrics.

Before
midsem

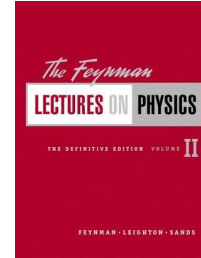
- **Magnetostatics:** Lorentz force, Biot-Savart and Ampere's laws and their applications, Divergence and Curl of Magnetostatic fields, Magnetic vector Potential, Force and torque on a magnetic dipole, Magnetic materials, Magnetization, Bound currents, Boundary conditions.
- **Electrodynamics:** Ohm's law, Motional EMF, Faraday's law, Lenz's law, Self and Mutual inductance, Energy stored in magnetic field, Maxwell's equations, Continuity Equation, Poynting Theorem, Wave solution of Maxwell Equations.
- **Electromagnetic waves:** Polarization, reflection & transmission at oblique incidences.

After
midsem

Text and References

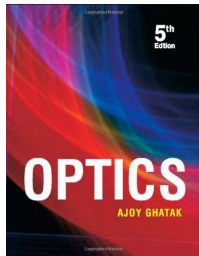


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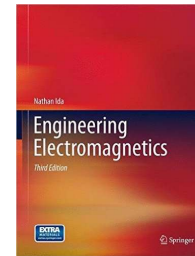


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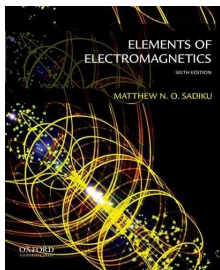
<http://www.feynmanlectures.caltech.edu/>



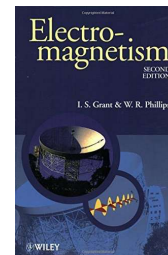
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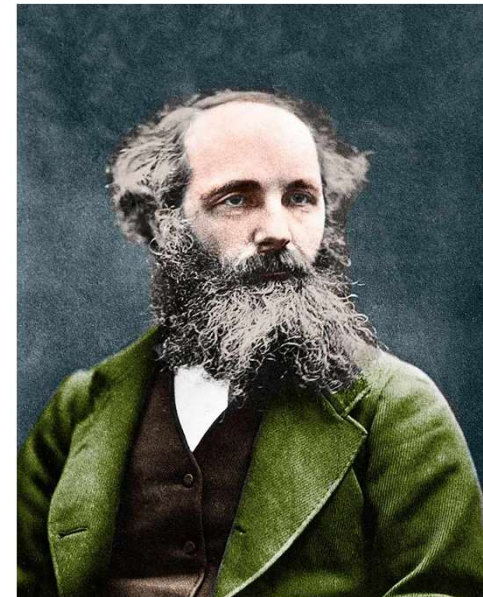
Electrodynamics

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho \quad (\text{Gauss's law}),$$

$$\nabla \cdot \mathbf{B} = 0 \quad (\text{no name}),$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (\text{Faraday's law}),$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (\text{Ampère's law with Maxwell's correction}).$$



James Clerk Maxwell
(13 June 1831 – 5 November 1879)

Maxwell's equations appeared in a paper published in 1864: "***A Dynamical Theory of the Electromagnetic Field***". These were further addressed in his ***Treatise on Electricity and Magnetism***, published in 1873.

What is *field*?

A physical quantity has a value in a region of space. The term **field** describes both, **the region and the value of the physical quantity in that region.**

A field is a distribution in space of any physical quantity.

Physical quantity

- Scalar or vector in nature.
- Time dependent or time independent.

Why concept of “field”?

- A function may or may not have a physical meaning. To distinguish meaningful functions, we introduce the concept of fields.
- What is the means of interaction between two objects? It helps in visualization or gives a “**feel**” of what is happening between two objects.

Objects in the context of electrodynamics could be charges or magnets.

Scalar field

- Each point in space is associated with a real number.
- May or may not be a function of time.

Examples:

Temperature, Pressure, Electric potential, Charge density

Representation:

A scalar field f as a function of position $\mathbf{r}$ and time t is expressed as

$$f(\mathbf{r}, t) = f(x, y, z, t)$$

where $\mathbf{r} = \hat{\mathbf{i}}x + \hat{\mathbf{j}}y + \hat{\mathbf{k}}z$

Vector field

- Each point in space is associated with a vector.
- May or may not be a function of time.

Examples:

Force, velocity, electric field.

Representation:

A vector field $\mathbf{v}$ as a function of position $\mathbf{r}$ and time t is expressed as

$$\mathbf{v}(\mathbf{r}, t) = \hat{\mathbf{i}}v_x(x, y, z, t) + \hat{\mathbf{j}}v_y(x, y, z, t) + \hat{\mathbf{k}}v_z(x, y, z, t)$$

Vector calculus

➤ Differential calculus

- Gradient (scalar field)
- Divergence (vector field)
- Curl (vector field)

➤ Integral calculus

- Line integral
- Surface integral
- Volume integral

➤ Fundamental theorems

- Gradient theorem
- Divergence theorem (Gauss's or Green's theorem)
- Curl theorem (Stokes theorem)

Vector Operations (Recap)

(i) **Addition of two vectors:** Place the tail of **B** at the head of **A**.

Commutative: $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$

Associative: $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$

(ii) **Multiplication by a scalar:**

Multiplies the magnitude but leaves the direction unchanged.

Distributive: $a(\mathbf{A} + \mathbf{B}) = a\mathbf{A} + a\mathbf{B}$

(iii) **Dot product of two vector (scalar product):**

The dot product of two vectors is defined by $\mathbf{A} \cdot \mathbf{B} = AB \cos \theta$,
and the result is scalar. Here θ is the angle they form when placed tail-to-tail.

Commutative: $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$

Distributive: $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$

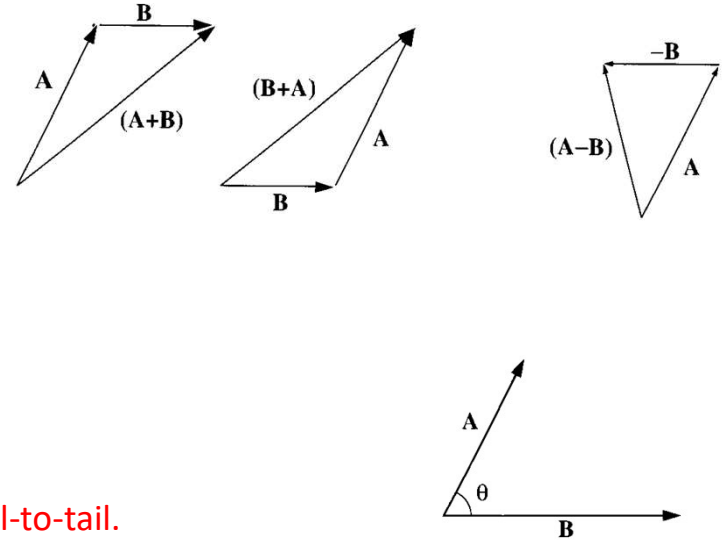
Geometrically $\mathbf{A} \cdot \mathbf{B}$ is the product of A and the component of B along A .

(iv) **Cross product of two vector (vector product):**

The cross product of two vectors is defined by, $\mathbf{A} \times \mathbf{B} = AB \sin \theta \hat{n}$ where $\hat{n}$ is a unit vector pointing perpendicular to the plane of **A** and **B**.

A hat is used to designate the unit vector and its direction is determined by the right-hand rule.

not commutative: $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$; *Distributive:* $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$



Vector Algebra: Component form (Recap)

Let $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ be unit vectors parallel to the x, y, and z axes, respectively.

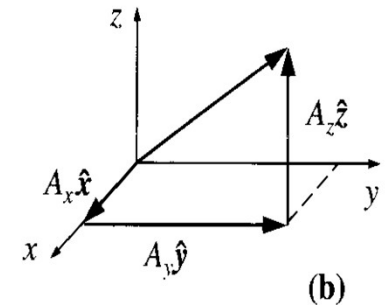
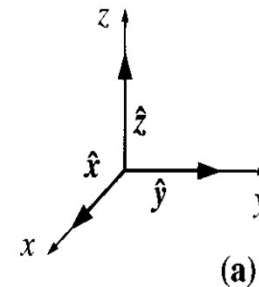
An arbitrary vector $\mathbf{A}$ can be expressed in terms of these basis vectors.

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}$$

The numbers A_x , A_y , and A_z are called components of $\mathbf{A}$.

(i) To add vectors, add like components.

$$\begin{aligned}\mathbf{A} + \mathbf{B} &= (A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}) + (B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}}) \\ &= (A_x + B_x) \hat{\mathbf{x}} + (A_y + B_y) \hat{\mathbf{y}} + (A_z + B_z) \hat{\mathbf{z}}.\end{aligned}$$



(ii) To multiply by a scalar, multiply each component.

$$a\mathbf{A} = (aA_x) \hat{\mathbf{x}} + (aA_y) \hat{\mathbf{y}} + (aA_z) \hat{\mathbf{z}}.$$

(iii) To calculate the dot product, multiply like components, and add.

$$\begin{aligned}\mathbf{A} \cdot \mathbf{B} &= (A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}) \cdot (B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}}) \\ &= A_x B_x + A_y B_y + A_z B_z.\end{aligned}$$

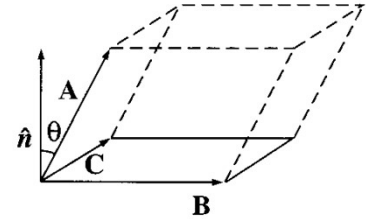
(iv) To calculate the cross product, form the determinant whose first row is $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ whose second row is $\mathbf{A}$ (in component form), and whose third row is $\mathbf{B}$.

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= (A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}) \times (B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}}) \\ &= (A_y B_z - A_z B_y) \hat{\mathbf{x}} + (A_z B_x - A_x B_z) \hat{\mathbf{y}} + (A_x B_y - A_y B_x) \hat{\mathbf{z}} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}.\end{aligned}$$

Triple product (Recap)

Since the cross product of two vectors is itself a vector, it can be dotted or crossed with a third vector to form a triple product.

(i) **Scalar triple product: $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$.** Geometrically,



$|\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})|$ is the volume of a parallelepiped generated by these three vectors as shown in Fig.

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

(ii) **Vector triple product: $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$.** The vector triple product can be simplified by the so-called **BAC-CAB** rule. $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$

Notice that $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} \neq \mathbf{A} \times (\mathbf{B} \times \mathbf{C})$

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = -\mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = -\mathbf{A}(\mathbf{B} \cdot \mathbf{C}) + \mathbf{B}(\mathbf{A} \cdot \mathbf{C})$$

Position, Displacement, and Separation Vectors (Recap)

Position vector: The vector that points from the origin.

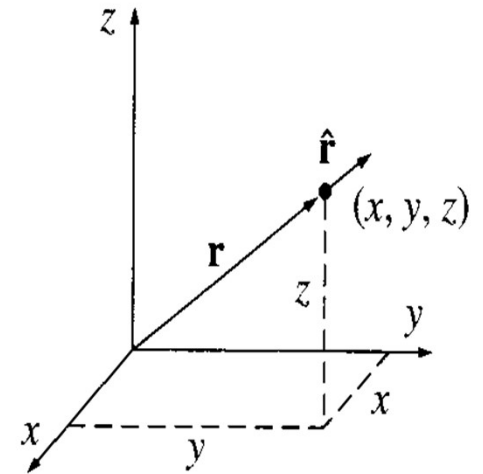
$$\mathbf{r} \equiv x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}$$

Its magnitude (the distance from the origin) is

$$r = \sqrt{\mathbf{r} \cdot \mathbf{r}} \equiv \sqrt{x^2 + y^2 + z^2}$$

Its direction is represented by a unit vector, $\hat{\mathbf{r}}$ (pointing radially outward)

$$\hat{\mathbf{r}} = \frac{\mathbf{r}}{r} = \frac{x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}}{\sqrt{x^2 + y^2 + z^2}}$$



The infinitesimal displacement vector, from (x, y, z) to $(x+dx, y+dy, z+dz)$, is

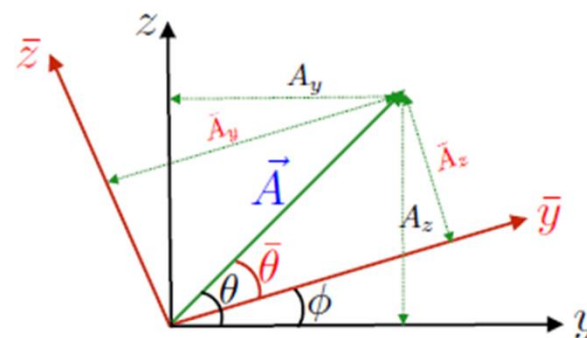
$$d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}.$$

Vector transformation (Recap)

Components of a vector has to transform “properly” under a coordinate change.

Suppose $\bar{x}, \bar{y}, \bar{z}$ coordinate system is rotated by angle ϕ relative to x, y, z about a common axis $x = \bar{x}$ which is perpendicular to the page.

$$A_y = A \cos \theta, \quad A_z = A \sin \theta$$



$$\bar{A}_y = A \cos \bar{\theta} = A \cos(\theta - \phi) = A(\cos \theta \cos \phi + \sin \theta \sin \phi) = \cos \phi A_y + \sin \phi A_z$$

$$\bar{A}_z = A \sin \bar{\theta} = A \sin(\theta - \phi) = A(\sin \theta \cos \phi - \cos \theta \sin \phi) = -\sin \phi A_y + \cos \phi A_z$$

In a compact form

$$\begin{pmatrix} \bar{A}_y \\ \bar{A}_z \end{pmatrix} = \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} A_y \\ A_z \end{pmatrix}$$

A scalar remains invariant under a change in coord. system

In general, for rotation about arbitrary axis in 3-D

$$\begin{pmatrix} \bar{A}_x \\ \bar{A}_y \\ \bar{A}_z \end{pmatrix} = \begin{pmatrix} R_{xx} & R_{xy} & R_{xz} \\ R_{yx} & R_{yy} & R_{yz} \\ R_{zx} & R_{zy} & R_{zz} \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} \Rightarrow \bar{A}_i = \sum_{j=1}^3 R_{ij} A_j$$

A vector is any set of 3 components that has the above transformation property

Representing Source and Field point

In electrodynamics one frequently encounters problems involving two points:

A source point, $\mathbf{r}'$, where an electric charge is located.

A field point, $\mathbf{r}$, at which the electric field is calculated.

A short-hand notation for the separation vector from the source point to the field

$$\mathbf{r} \equiv \mathbf{r} - \mathbf{r}'.$$

Its magnitude is

$$r = |\mathbf{r} - \mathbf{r}'|,$$

and a unit vector in the direction from $\mathbf{r}'$ to $\mathbf{r}$ is

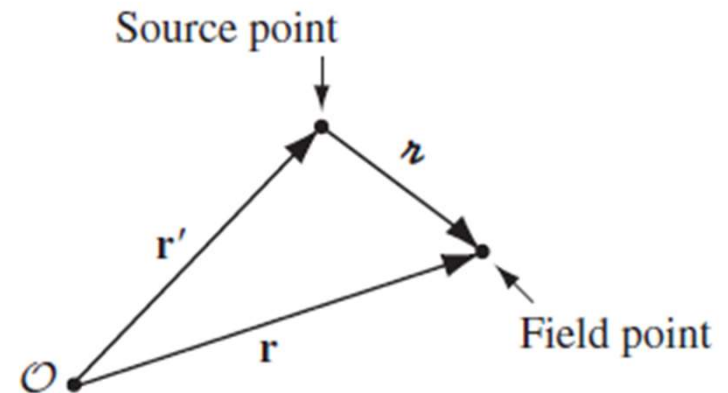
$$\hat{\mathbf{r}} = \frac{\mathbf{r}}{r} = \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|}.$$

In Cartesian coordinates,

$$\mathbf{r} = (x - x')\hat{\mathbf{x}} + (y - y')\hat{\mathbf{y}} + (z - z')\hat{\mathbf{z}},$$

$$r = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2},$$

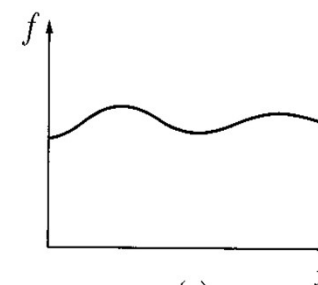
$$\hat{\mathbf{r}} = \frac{(x - x')\hat{\mathbf{x}} + (y - y')\hat{\mathbf{y}} + (z - z')\hat{\mathbf{z}}}{\sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}}$$



Differential Calculus

“Ordinary” Derivatives

Suppose we have a function of one variable: $f(x)$. The derivative, df/dx , tells us how rapidly the function $f(x)$ varies when we change the argument x by a tiny amount, dx :



$$df = \left(\frac{df}{dx} \right) dx.$$

In words, if we change x by an amount dx , then, f changes by an amount df . The derivative is a proportionality constant and it describes how rapidly the given function varies.

df/dx is small for f varying slowly with x ; it would be large if the variation was rapid.

Geometrical interpretation: The derivative df/dx is the slope of the graph of f versus x .

Gradient

Suppose we have a function of three variables $T(x, y, z)$. A theorem on partial derivatives states that

if $x \longrightarrow x$ to $x + dx$, $y \longrightarrow y$ to $y + dy$, $z \longrightarrow z$ to $z + dz$

then a differential change in $T(x, y, z)$ can be written as

$$\begin{aligned}dT &= \frac{\partial T}{\partial x} dx + \frac{\partial T}{\partial y} dy + \frac{\partial T}{\partial z} dz \\&= (\hat{\mathbf{i}} \frac{\partial T}{\partial x} + \hat{\mathbf{j}} \frac{\partial T}{\partial y} + \hat{\mathbf{k}} \frac{\partial T}{\partial z}) \cdot (\hat{\mathbf{i}} dx + \hat{\mathbf{j}} dy + \hat{\mathbf{k}} dz) \\&= \vec{\nabla} T \cdot d\mathbf{r}\end{aligned}$$

where

$$\vec{\nabla} T = (\hat{\mathbf{i}} \frac{\partial T}{\partial x} + \hat{\mathbf{j}} \frac{\partial T}{\partial y} + \hat{\mathbf{k}} \frac{\partial T}{\partial z})$$

is the gradient of the scalar function $T(x, y, z)$.

Geometrical interpretation of gradient

We write

$$\begin{aligned} dT &= \vec{\nabla} T \cdot d\mathbf{r} \\ &= |\vec{\nabla} T| |d\mathbf{r}| \cos \theta \end{aligned}$$

Maximum change in T occurs at $\theta = 0$ for a fixed $|d\mathbf{r}|$.

$\vec{\nabla} T$ points in the direction of maximum increase of T .

$|\vec{\nabla} T|$ is the rate of increase of T along this maximal direction.

$\vec{\nabla}$ is a vector differential operator. All alone, it has no meaning.

Imagine you are standing on a hill side. Look all around you and find the direction of steepest ascent. That is the direction of gradient and the slope in that direction is its magnitude.

Directional derivative

We can also write

$$\begin{aligned}dT &= \vec{\nabla} T \cdot d\mathbf{r} \\ &= \vec{\nabla} T \cdot \hat{n} dl\end{aligned}$$

$$\frac{dT}{dl} = \vec{\nabla} T \cdot \hat{n}$$

dT/dl $\longrightarrow$ **directional derivative** i.e, rate of change of T with respect to distance l in a directional other than that along which ∇T is maximal.

The Operator ∇

The gradient has the formal appearance of a vector, ∇ , “multiplying”, a scalar T .

$$\nabla T = \left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right) T.$$

∇ is a **vector operator** that acts upon T and not a vector that multiplies T .

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}.$$

∇ mimics the behavior of an ordinary vector in virtually every way, if we translate “multiply” by “act upon”. It is a marvelous piece of notational simplification

An ordinary vector $\mathbf{A}$ can multiply in three ways:

1. Multiply a scalar a : $a\mathbf{A}$
2. Multiply another vector (dot product): $\mathbf{A} \cdot \mathbf{B}$
3. Multiply another vector (cross product): $\mathbf{A} \times \mathbf{B}$

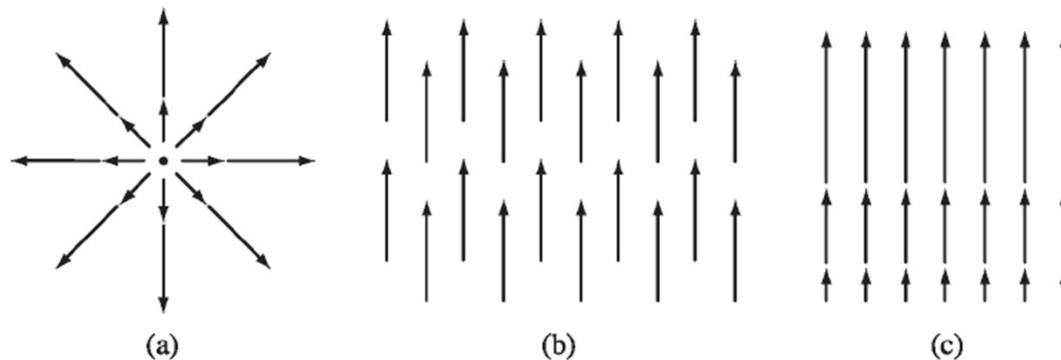
Correspondingly, there are three ways the operator ∇ can act:

1. On a scalar function T : ∇T (**Gradient**)
2. On a vector function (dot product): $\nabla \cdot \mathbf{v}$ (**divergence**)
3. On a vector function (cross product): $\nabla \times \mathbf{v}$ (**curl**)

The Divergence

Divergence of a vector $\mathbf{v}$ is:
$$\nabla \cdot \mathbf{v} = \left(\hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \right) \cdot (v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} + v_z \hat{\mathbf{z}})$$
$$= \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}.$$

$\nabla \cdot \mathbf{v}$ is a measure of how much the vector $\mathbf{v}$ spread out from the point in question.
(divergence of a scalar is meaningless).



Ex: Suppose the functions in the above figures $\mathbf{v}_a = \mathbf{r} = x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}$, $\mathbf{v}_b = \hat{\mathbf{z}}$, and $\mathbf{v}_c = z \hat{\mathbf{z}}$.
Calculate their divergences.

$$\nabla \cdot \mathbf{v}_a = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3.$$

$$\nabla \cdot \mathbf{v}_b = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(1) = 0 + 0 + 0 = 0,$$

$$\nabla \cdot \mathbf{v}_c = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(z) = 0 + 0 + 1 = 1.$$

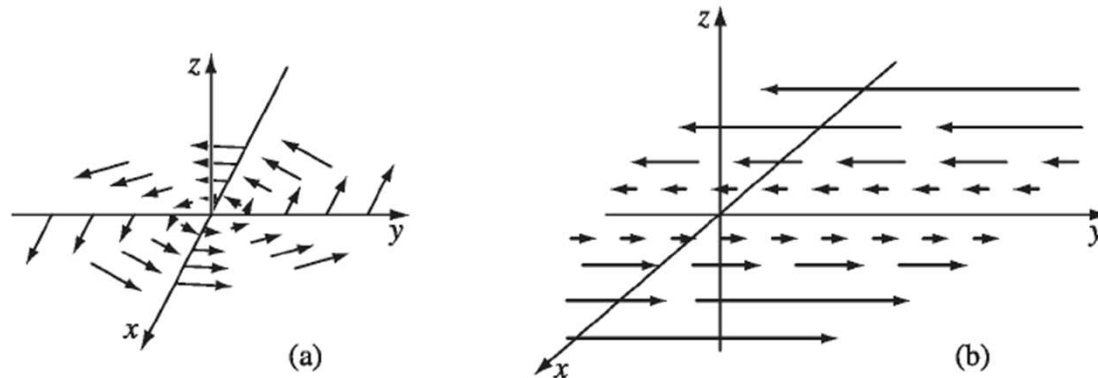
The curl

The curl of a vector is

$$\nabla \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_x & v_y & v_z \end{vmatrix}$$

$$= \hat{\mathbf{x}} \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) + \hat{\mathbf{y}} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) + \hat{\mathbf{z}} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right).$$

$\nabla \times \mathbf{v}$ is a measure of how much the vector $\mathbf{v}$ curl around the point in question.



Suppose the functions in above two figures $\mathbf{v}_a = -y\hat{\mathbf{x}} + x\hat{\mathbf{y}}$, $\mathbf{v}_b = x\hat{\mathbf{y}}$.
 Calculate their curls.

$$\nabla \times \mathbf{v}_a = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -y & x & 0 \end{vmatrix} = 2\hat{\mathbf{z}},$$

$$\nabla \times \mathbf{v}_b = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 0 & x & 0 \end{vmatrix} = \hat{\mathbf{z}}.$$

Product rules

The calculation of ordinary derivatives is facilitated by a number of rules, such as the sum rule:

$$\frac{d}{dx}(f + g) = \frac{df}{dx} + \frac{dg}{dx},$$

the rule for multiplying by a constant:

$$\frac{d}{dx}(kf) = k \frac{df}{dx},$$

the product rule:

$$\frac{d}{dx}(fg) = f \frac{dg}{dx} + g \frac{df}{dx},$$

and the quotient rule:

$$\frac{d}{dx} \left(\frac{f}{g} \right) = \frac{g \frac{df}{dx} - f \frac{dg}{dx}}{g^2}.$$

Similar relations hold for the vector derivatives. Thus,

$$\nabla(f + g) = \nabla f + \nabla g, \quad \nabla \cdot (\mathbf{A} + \mathbf{B}) = (\nabla \cdot \mathbf{A}) + (\nabla \cdot \mathbf{B}),$$

$$\nabla \times (\mathbf{A} + \mathbf{B}) = (\nabla \times \mathbf{A}) + (\nabla \times \mathbf{B}),$$

and

$$\nabla(kf) = k \nabla f, \quad \nabla \cdot (k\mathbf{A}) = k(\nabla \cdot \mathbf{A}), \quad \nabla \times (k\mathbf{A}) = k(\nabla \times \mathbf{A}),$$

Product rules for the gradients

$$(i) \quad \nabla(fg) = f\nabla g + g\nabla f,$$

$$(ii) \quad \nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A},$$

two for divergences:

$$(iii) \quad \nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f),$$

$$(iv) \quad \nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}),$$

and two for curls:

$$(v) \quad \nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f),$$

$$(vi) \quad \nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}).$$

Following quotient rules can be easily obtained from the corresponding product rules:

$$\nabla \left(\frac{f}{g} \right) = \frac{g\nabla f - f\nabla g}{g^2}, \quad \nabla \cdot \left(\frac{\mathbf{A}}{g} \right) = \frac{g(\nabla \cdot \mathbf{A}) - \mathbf{A} \cdot (\nabla g)}{g^2}, \quad \nabla \times \left(\frac{\mathbf{A}}{g} \right) = \frac{g(\nabla \times \mathbf{A}) + \mathbf{A} \times (\nabla g)}{g^2}.$$

Caution !!!

For usual vectors that we use, say, $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, the vector identity is

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}).$$

Suppose we replace $\mathbf{A}$ by ∇ in the above vector identity, then we have

$$\nabla \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\nabla \cdot \mathbf{C}) - \mathbf{C}(\nabla \cdot \mathbf{B}).$$

However, the correct identity is

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}).$$

This indicates that ∇ as an operator does not follow the rules of usual vector identities. Mere replacement of an operator does not reproduce the vector identity.

Solved example 1

Find the gradient of $r = \sqrt{x^2 + y^2 + z^2}$ (the magnitude of the position vector).

$$\begin{aligned}\nabla r &= \frac{\partial r}{\partial x} \hat{\mathbf{x}} + \frac{\partial r}{\partial y} \hat{\mathbf{y}} + \frac{\partial r}{\partial z} \hat{\mathbf{z}} \\ &= \frac{1}{2} \frac{2x}{\sqrt{x^2 + y^2 + z^2}} \hat{\mathbf{x}} + \frac{1}{2} \frac{2y}{\sqrt{x^2 + y^2 + z^2}} \hat{\mathbf{y}} + \frac{1}{2} \frac{2z}{\sqrt{x^2 + y^2 + z^2}} \hat{\mathbf{z}} \\ &= \frac{x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}}{\sqrt{x^2 + y^2 + z^2}} = \frac{\mathbf{r}}{r} = \hat{\mathbf{r}}.\end{aligned}$$

Conclusion: the distance from the origin increases most rapidly in the radial direction, and that its *rate* of increase in that direction is 1. **(Check with the definition of gradient)**

Solved example 2

Suppose the height of the hill (in feet) behind the CC is $h(x, y) = 10(2xy - 3x^2 - 4y^2 - 18x + 28y + 12)$, where y is the distance (in miles) north, x the distance west of the IITG admin. building. Where is the top of the hill and how high is the hill? How steep is the slope 1 mile north and 1 mile west of admin building?



$$\begin{aligned}\vec{\nabla}h(x, y) &= \frac{\partial h}{\partial x}\hat{x} + \frac{\partial h}{\partial y}\hat{y} \\ &= 10(2y - 6x - 18)\hat{x} + 10(2x - 8y + 28)\hat{y}\end{aligned}$$

If the hilltop is located at $(\bar{x}, \bar{y})$, then $\vec{\nabla}h|_{(\bar{x}, \bar{y})} = 0$
(condition for maxima)

$$\begin{aligned}\text{This gives} \quad \begin{aligned} 2\bar{y} - 6\bar{x} - 18 &= 0 \\ 2\bar{x} - 8\bar{y} + 28 &= 0. \end{aligned} &\implies \bar{x} = -2 \text{ and } \bar{y} = 3\end{aligned}$$

The hill top is 3 miles north and 2 miles east of the admin building.

Height $h(-2, 3) = 720$ feet.

The slope at a point 1 mile west and 1 mile north of admin building will be found from the magnitude of $\vec{\nabla}h(x, y)$ at $x = 1, y = 1$, which gives $\vec{\nabla}h(1, 1) = 220(-\hat{x} + \hat{y})$. Slope $\implies 220\sqrt{2}$ ft/mile.

Questions?